

Optical Satellite Data

Copernicus, Sentinel and the current EO landscape

Access • vegetation monitoring • complementary missions
• cloud platforms • GeoAI

Katarzyna Ostapowicz

Norwegian Institute for Nature Research (NINA)
katarzyna.ostapowicz@nina.no



https://www.esa.int/ESA_Multimedia/Images/2025/05/Earth_from_Space_Svalbard_Archipelago

A practical map of the current EO landscape

Enough context to make good choices — without turning the first hour into a catalogue of missions and platforms.

1

Copernicus first

Sentinel missions, CDSE and Copernicus services

2

Then complement

Landsat, JAXA, Korean missions and commercial constellations

3

Then compute

Colab + APIs today; GEE, WEkEO and GeoAI as context

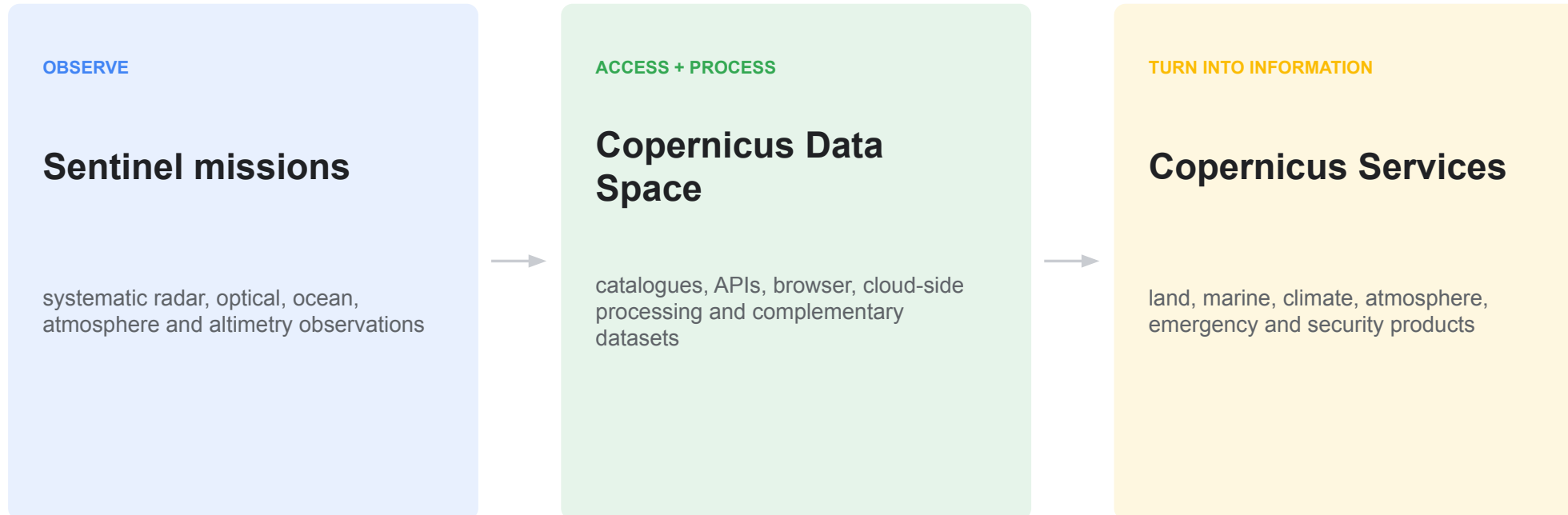
≈20 min lecture

≈40 min practical

The practical remains one workflow: Sentinel-2 L2A for Adventdalen.

Copernicus is an ecosystem, not only a satellite archive

Observations, a data space and thematic services form one workflow from measurement to environmental information.



For vegetation research, the useful chain is often: Sentinel-2 / Sentinel-3 → CDSE → CLMS / your own analysis.

Sentinel-2 and vegetation

High spatial detail, red-edge bands and systematic acquisitions make Sentinel-2 a strong default for vegetation mapping.

Why it matters

10 m	visible + NIR detail
20 m	red-edge + SWIR information
L2A	surface reflectance + scene classification
systematic	repeat coverage for seasonal monitoring

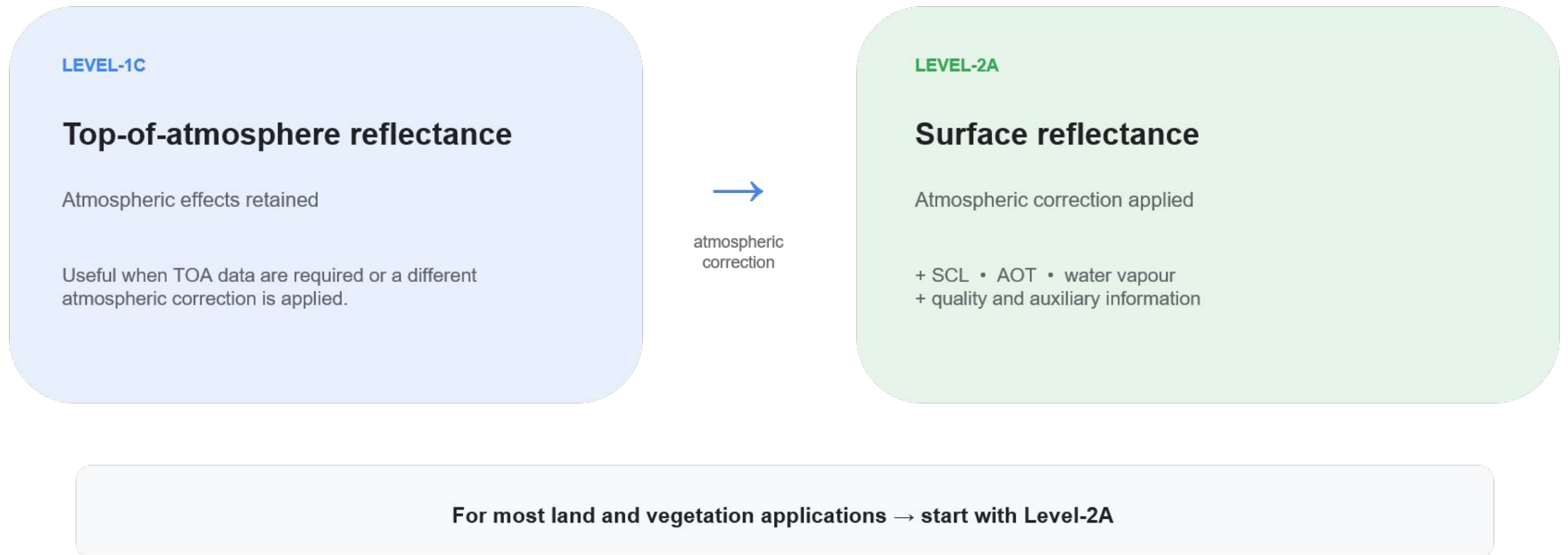
A useful subset for this course



Arctic caution

Cloud, snow, shadows and low illumination can dominate the usable-observation budget.

The processing level determines what has already been corrected — and which quality layers are available.



Sentinel-2A, B and C: one mission, several satellites

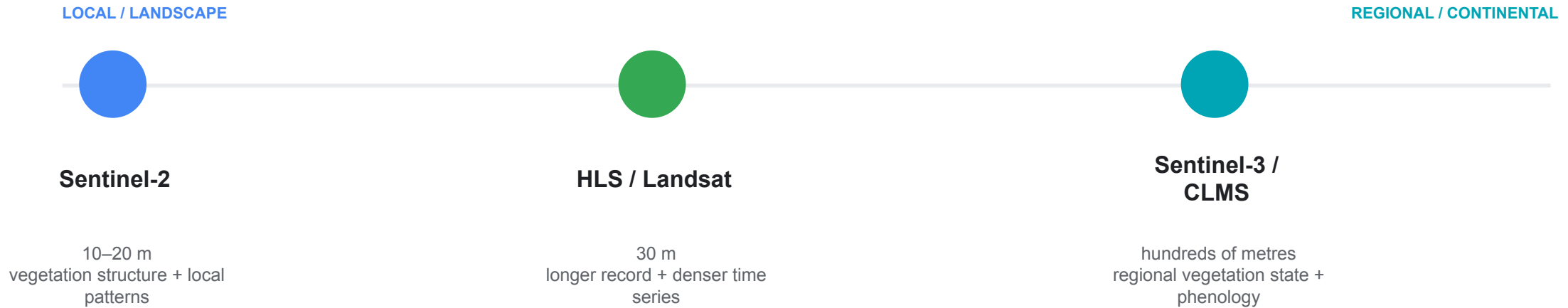
The letter identifies the spacecraft — not a different sensor or product type.



Sources: <https://sentiwiki.copernicus.eu/web/s2-mission> • <https://dataspace.copernicus.eu/news/2026-5-15-sentinel-2a-extension-campaign-prolonged-until-end-2026>

When the scale changes: Sentinel-3 and Copernicus Land

Fine spatial detail and broad-area temporal consistency answer different questions.

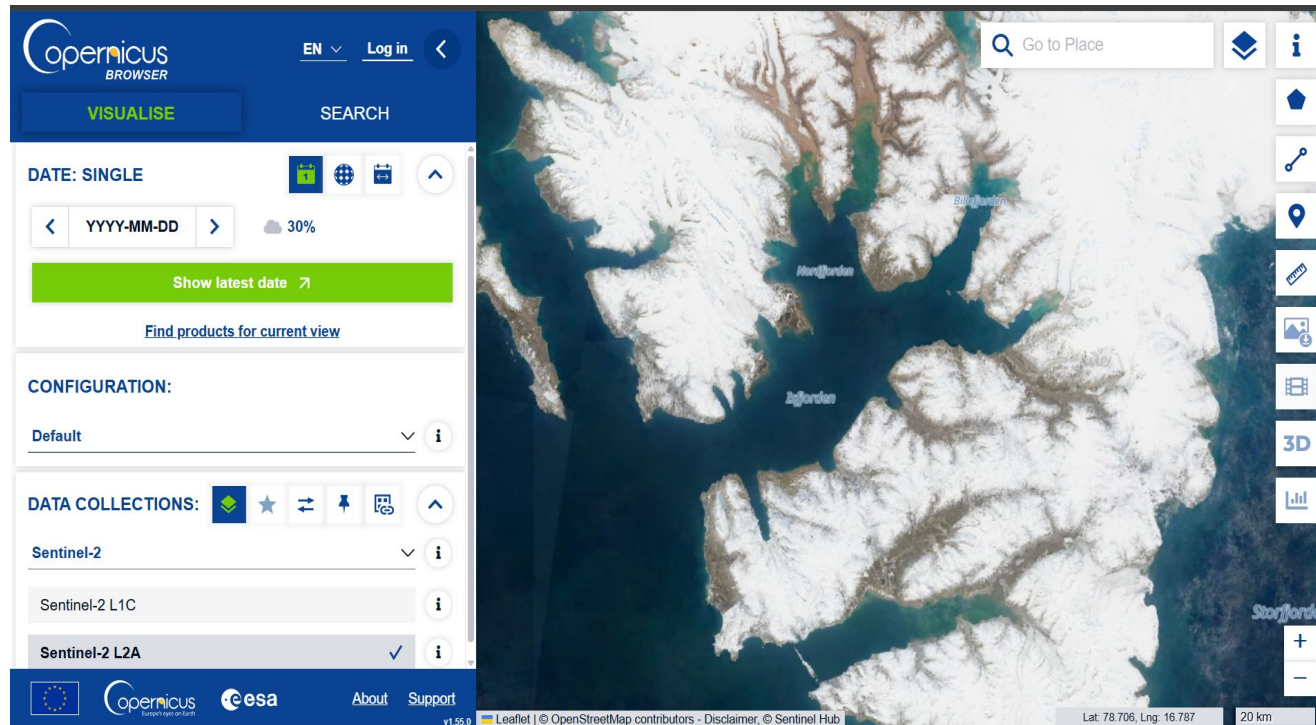


CLMS already provides variables such as NDVI, LAI, FAPAR, vegetation condition and land-surface phenology.

Use the finest resolution only when the question needs it.

Copernicus Data Space Ecosystem

One environment for discovery, catalogue access, data delivery and server-side processing.



<https://browser.dataspace.copernicus.eu/>

DISCOVER

Browser • STAC

ACCESS

S3 • OData • catalogue APIs

PROCESS

Process API • openEO •
JupyterLab

CDSE also includes Copernicus Contributing Missions and complementary / federated datasets.

Pixel quality: what can we use?

Atmospheric correction is only the start — cloud, shadow, snow and invalid pixels still need explicit screening.

Delivered with Sentinel-2 L2A / CDSE

SCL

cloud • cirrus • shadow • snow/ice • water • land

dataMask

source data / no data

CLD / SNW

cloud and snow probability

AOT / WVP

aerosol and water-vapour information

Additional screening approaches

s2cloudless

continuous cloud probability

Cloud Score+

continuous clear-pixel quality score

Spectral tests

blue / cirrus / SWIR can help diagnose cloud and snow

Temporal screening

compare neighbouring observations through time

Arctic caution: snow, bright surfaces, thin cloud and low illumination can challenge any single mask.

Source: <https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/S2L2A.html>

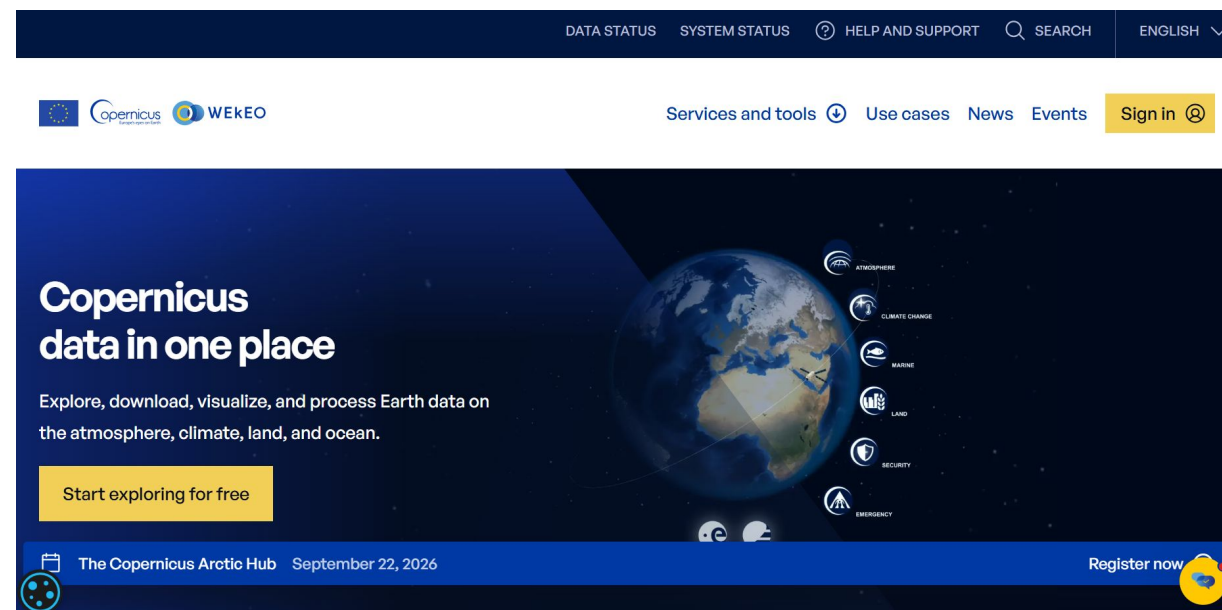
WEkEO: useful when the question crosses domains

A complementary route into land, climate, ocean and atmosphere data — especially useful for Arctic environmental context.

Good reasons to use it

- **Cross-service catalogue** Copernicus Land, Marine, Climate, Atmosphere and more
- **Environmental context** link optical observations to sea ice, climate and atmosphere
- **Cloud workspace** data viewer, API access and Jupyter-based processing

Not a replacement for CDSE — a different entry point for multi-domain environmental data.



<https://www.wekeo.eu/>

September 22nd - September 24th

The Copernicus Arctic Hub

Data and Tools for the Arctic

👤 Register now

ℹ More information

About the training

<https://events.wekeo.eu/the-copernicus-arctic-hub>

Beyond Copernicus: public missions that add something different

Choose them for temporal depth, spectral range, spatial detail or all-weather capability — not simply because they are another dataset.

US

Landsat 8 / 9 + HLS

time series

- long land-observation record
- thermal information
- HLS harmonises Landsat + Sentinel-2 at 30 m

JAPAN

GCOM-C / SGLI

regional optical

- 250 m / 1 km products
- vegetation, snow, ocean colour
- broad-area repeat observations

KOREA

KOMPSAT

high resolution

- sub-metre / metre-class optical missions
- local mapping and detailed reference
- different access model from Sentinel

CANADA

RCM

radar complement

- not optical
- all-weather C-band SAR
- very strong Arctic relevance

For vegetation, HLS is especially useful when temporal density matters; native Sentinel-2 remains preferable when 10 m detail or red-edge bands matter.

Commercial imagery fills gaps in spatial detail and revisit

Open Sentinel and Landsat data remain the backbone; commercial constellations become valuable when the research question needs more detail or frequency.

PLANET

PlanetScope

≈3 m • near-daily

Analysis-ready surface-reflectance products are available for dense time series; SkySat provides finer tasking.

AIRBUS

Pléiades Neo

30 cm

Very-high-resolution optical imagery for detailed mapping, validation and small-area change analysis.

VANTOR

WorldView

30 cm class

Very-high-resolution optical imagery, including multispectral products for detailed land and vegetation applications.

Examples, not a ranking. Cost, licensing, archive depth, spectral configuration and tasking requirements can matter as much as nominal resolution.

How ESA and Copernicus bring other missions into the workflow

The European EO landscape deliberately combines Sentinel observations with national, international and commercial data.

COPERNICUS CONTRIBUTING MISSIONS

Data that complement Sentinel observations for Copernicus needs

Example: PlanetScope is available in CDSE within Copernicus Contributing Mission datasets.

ESA THIRD PARTY MISSIONS

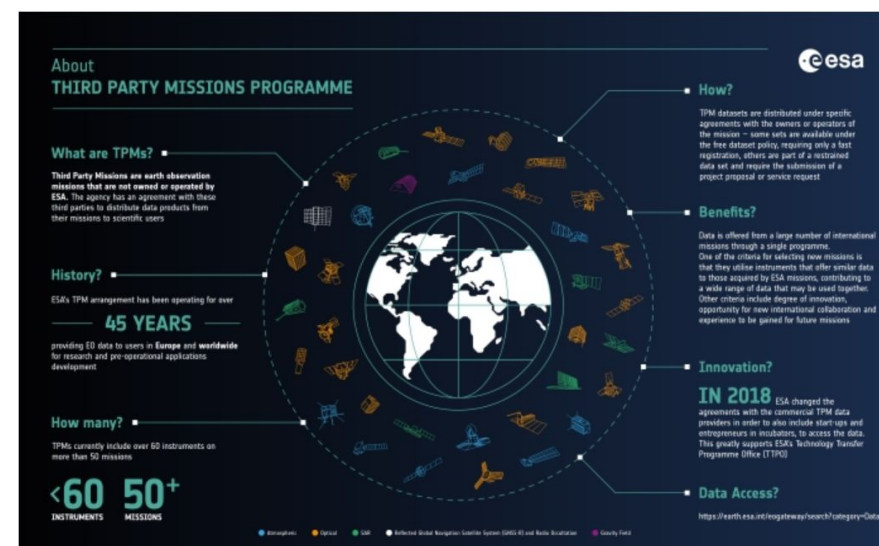
ESA agreements to distribute data from missions it does not own or operate

International and commercial missions; access conditions vary by mission and user category.

ESA NETWORK OF RESOURCES

Cloud and platform services for EO exploitation

Useful when processing needs are larger than a local notebook or desktop workflow.



<https://earth.esa.int/eogateway/missions/third-party-missions>

Availability through ESA/Copernicus does not automatically mean unrestricted open access.

The same data can support different compute models

Choose the environment for the scale and workflow — not because one platform is universally “better”.

1

Colab + CDSE APIs

transparent Python
small teaching workflow
our choice today

2

Google Earth Engine

large catalogue
server-side analysis
fast prototyping

3

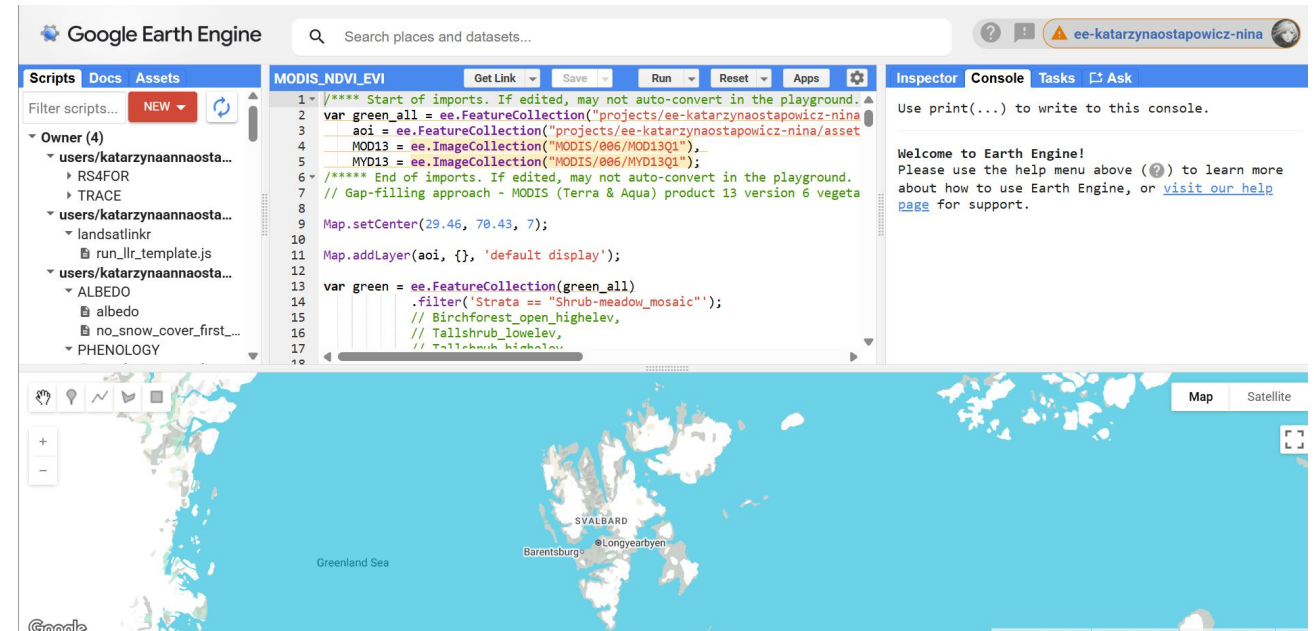
CDSE Jupyter / openEO

compute close to data
standards-based workflows
scalable processing

4

WEkEO workspace

cross-service data
cloud notebooks
environmental workflows



<https://code.earthengine.google.com/>

For this course:

one environment is enough. Google Colab runs the Python; CDSE provides the data and APIs.

GeoAI in 2026: foundation models and satellite embeddings

The shift is from using only raw bands and indices to also reusing representations learned from very large EO archives.

Prithvi-EO-2.0

NASA + IBM

pre-trained on global HLS time series

TerraMind

ESA Φ -lab + IBM

multimodal Earth-observation foundation model

AlphaEarth

Google DeepMind

annual 10 m satellite embeddings in Earth Engine

CDSE embeddings

Copernicus Data Space

large complementary embedding datasets from Sentinel archives

What changes: features can be reused across tasks

What does not: quality control, labels, scale and scientific interpretation still matter

THOR: a Norwegian geospatial foundation model

A compute-adaptive model designed to learn across Copernicus sensors, spatial resolutions and downstream tasks.

One model — heterogeneous observations



Why relevant here?

S1 + S2 + S3

Arctic multi-sensor context

limited labels

embeddings for GeoAI

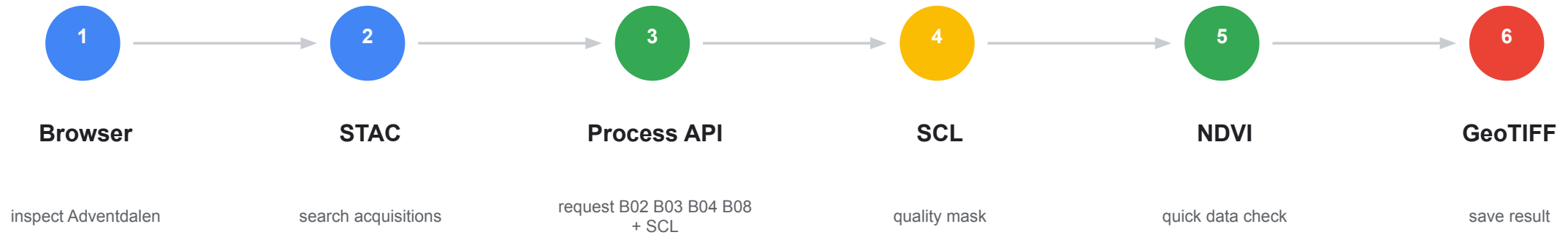
The screenshot shows the GitHub repository for THOR. The repository is public and has 3 branches and 1 tag. The main branch is selected. The repository contains several files and folders, including notebooks, tests, and thor. The repository was created 10 months ago and has 25 commits. The repository is managed by tforgaard.

File/Folder	Changes	Time
notebooks	Changes:	10 months ago
tests	add note to readme, update tests	5 months ago
thor	fallback to default patch size for channels missing in flexivit...	5 months ago
.gitignore	init THOR repo	10 months ago
LICENSE	init THOR repo	10 months ago
README.md	add note to readme, update tests	5 months ago
main.py	Changes:	9 months ago
pyproject.toml	less verbose logging during init, dtype casting fix, don't forc...	5 months ago

Example: https://github.com/FM4CS/thor_terratorch_ext/blob/main/notebooks/thor_embedding_generation_sen1ffloods11.ipynb

What we actually do today

One small Sentinel-2 L2A workflow in Google Colab. The wider landscape above is context for later choices.



Python environment

Google Colab

runs the code

Data + services

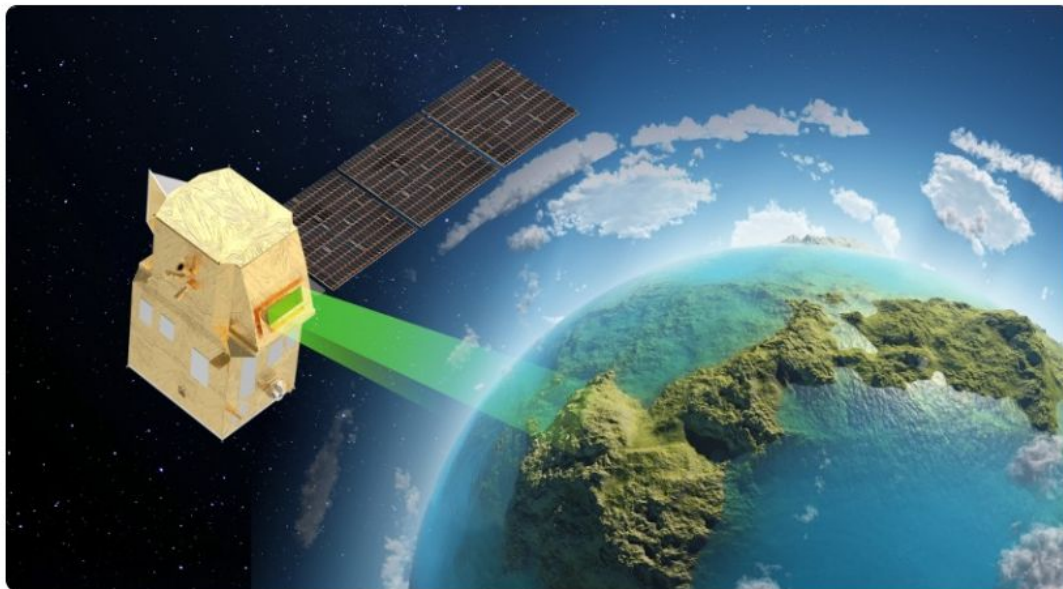
Copernicus Data Space

serves the data

The purpose is the access + preprocessing chain, not NDVI itself.

CHIME: from multispectral to hyperspectral Copernicus

The Sentinel Expansion mission will add dense spectral information for vegetation, soils, biodiversity and water applications.



What changes?

Sentinel-2



13 multispectral bands

CHIME



>200 narrow bands • 400–2500 nm

~30 m

spatial resolution

11 days

two-satellite revisit

L2A

surface reflectance

Complement Sentinel-2: less spatial detail, much richer spectral information.

FLEX: from reflectance to photosynthetic function

A new vegetation mission designed to measure solar-induced chlorophyll fluorescence rather than only surface reflectance.

ESA EARTH EXPLORER

Mission Overview

The FLuorescence EXplorer (FLEX) mission will provide global maps of vegetation fluorescence that can reflect photosynthetic activity and plant health and stress. In turn, this is not only important for a better understanding of the global carbon cycle, but also for agricultural management and food security. This is of particular relevance due to demand the growing world population is placing on the production of food.

Currently it is not possible to measure photosynthetic activity from space, but FLEX's novel instrument will be capable of achieving this. The mission was selected as the eighth Earth Explorer in ESA's Living Planet Programme in 2015.

FLEX will fly in tandem with the Copernicus Sentinel-3 mission, in particular working in combination with the OLCI and SLSTR instruments Sentinel-3 carries.

FLEX was launched on 15 September 2026, with a three and a half year design lifetime.



What is different?

FLORIS	imaging spectrometer
~300 m	global fluorescence mapping
SIF	information linked to photosynthetic activity
Sentinel-3	coordinated observations for atmospheric and surface context

Reflectance tells us how vegetation looks; fluorescence adds information on how photosynthesis is functioning.

https://www.esa.int/Applications/Observing_the_Earth/FutureEO/FLEX

References

- Copernicus Browser: <https://browser.dataspace.copernicus.eu/>
- CDSE STAC Browser: <https://browser.stac.dataspace.copernicus.eu/>
- CDSE documentation: <https://documentation.dataspace.copernicus.eu/>